

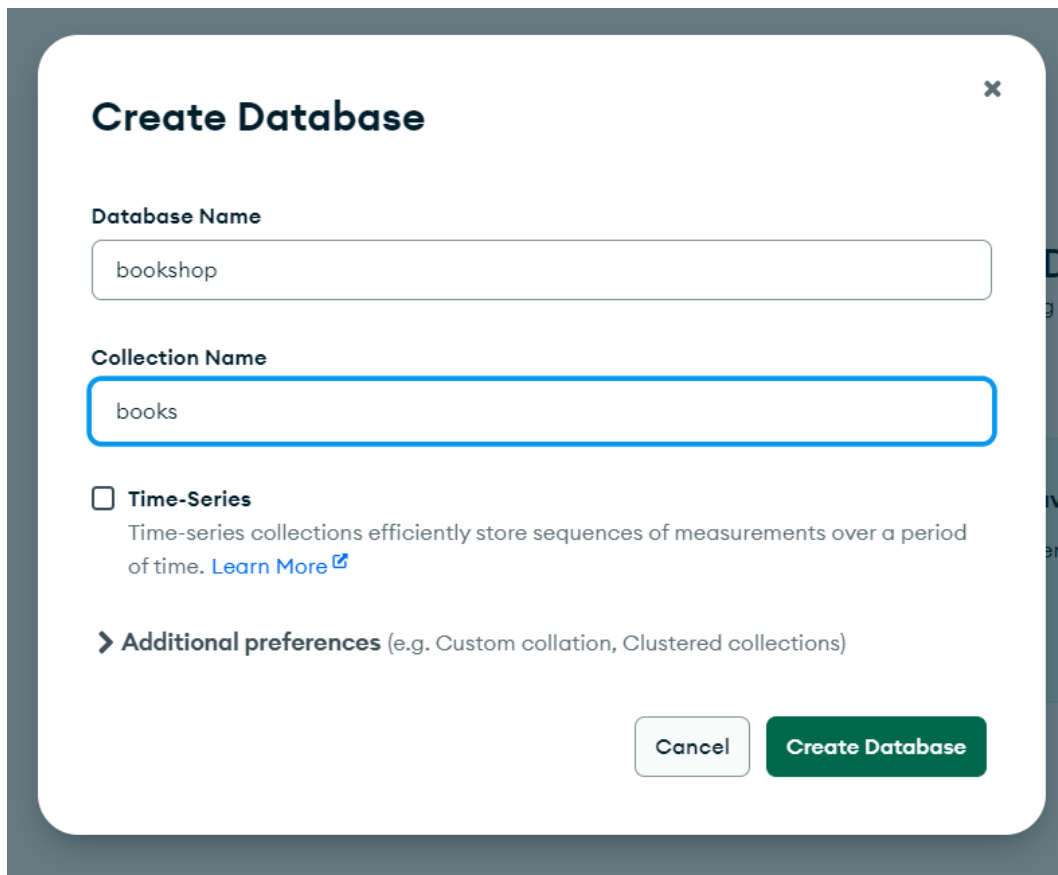
ASSIGNMENT 01

Registration No: 2021/ICT/35

Course Code: IT2234

Semester: II

1. Create a database bookshop
2. Create a collection books



Create Database ×

Database Name

bookshop

Collection Name

books

☐ **Time-Series**
Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

➤ **Additional preferences** (e.g. Custom collation, Clustered collections)

Cancel Create Database

3. Insert the following data

```
{  
  "title": "The Hobbit",  
  "author": "J.R.R. Tolkien",  
  "published_year": 1937,  
}
```

```

    "genres": ["Fantasy", "Adventure"],
    "pages": 310,
    "available": true,
    "rating": 4.8
  },
  {
    "title": "1984",
    "author": "George Orwell",
    "published_year": 1949,
    "genres": ["Dystopian", "Political Fiction"],
    "pages": 328,
    "available": true,
    "rating": 4.7
  },
  {
    "title": "To Kill a Mockingbird",
    "author": "Harper Lee",
    "published_year": 1960,
    "genres": ["Southern Gothic", "Drama"],
    "pages": 281,
    "available": false,
    "rating": 4.6
  },
  {
    "title": "The Great Gatsby",
    "author": "F. Scott Fitzgerald",
    "published_year": 1925,
    "genres": ["Tragedy"],
    "pages": 180,
    "available": true,
    "rating": 4.4
  },
  {
    "title": "Brave New World",
    "author": "Aldous Huxley",
    "published_year": 1932,
    "genres": ["Science Fiction", "Dystopian"],

```

```
"pages": 311,  
"available": true,  
"rating": 4.2  
}
```

Implementation:

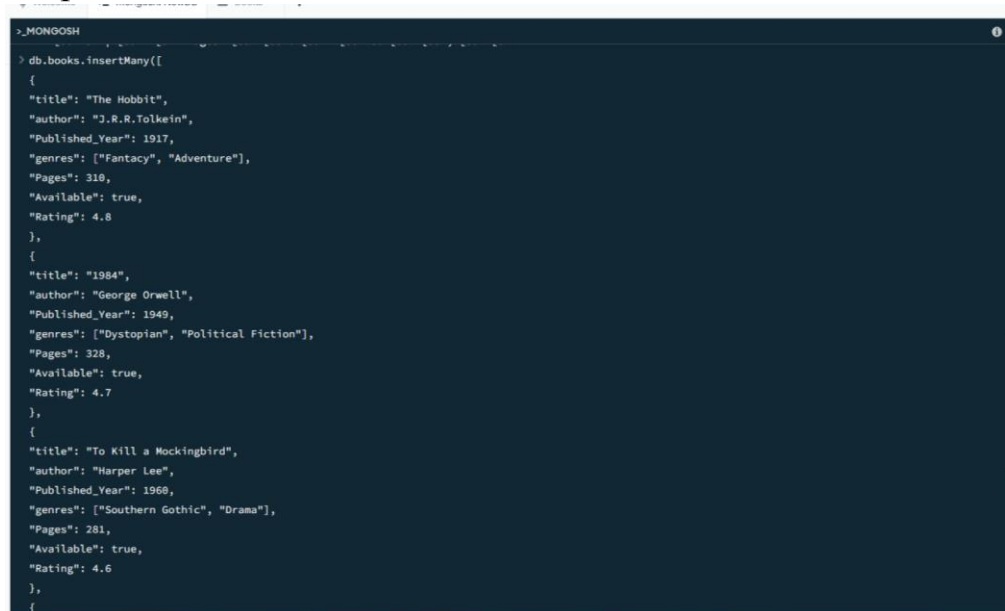
```
db.books.insertMany([  
  {  
    "title": "The Hobbit",  
    "author": "J.R.R. Tolkien",  
    "Published_year": 1937,  
    "genres": ["Fantasy", "Adventure"],  
    "pages": 310,  
    "Available": true,  
    "Rating": 4.8  
  },  
  {  
    "title": "1984",  
    "author": "George Orwell",  
    "Published_year": 1949,  
    "genres": ["Dystopian", "Political Fiction"],  
    "pages": 328,  
    "Available": true,  
    "Rating": 4.7  
  },  
  {  
    "title": "To Kill a Mockingbird",  
    "author": "Harper Lee",  
    "Published_year": 1960,  
    "genres": ["Southern Gothic", "Drama"],  
    "pages": 281,  
    "Available": false,  
    "Rating": 4.6  
  },  
  {
```

```

    "title": "The Great Gatsby",
    "author": "F. Scott Fitzgerald",
    "Published_year": 1925,
    "genres": ["Tragedy"],
    "pages": 180,
    "Available": true,
    "Rating": 4.4
  },
  {
    "title": "Brave New World",
    "author": "Aldous Huxley",
    "Published_year": 1932,
    "genres": ["Science Fiction", "Dystopian"],
    "pages": 311,
    "Available": true,
    "Rating": 4.2
  }
]

```

Output:



```

>_MONGODB
> db.books.insertMany([
  {
    "title": "The Hobbit",
    "author": "J.R.R.Tolkien",
    "Published_Year": 1917,
    "genres": ["Fantasy", "Adventure"],
    "Pages": 310,
    "Available": true,
    "Rating": 4.8
  },
  {
    "title": "1984",
    "author": "George Orwell",
    "Published_Year": 1949,
    "genres": ["Dystopian", "Political Fiction"],
    "Pages": 328,
    "Available": true,
    "Rating": 4.7
  },
  {
    "title": "To Kill a Mockingbird",
    "author": "Harper Lee",
    "Published_Year": 1960,
    "genres": ["Southern Gothic", "Drama"],
    "Pages": 281,
    "Available": true,
    "Rating": 4.6
  },
  {

```

```

>_MONGOSH
{
  "title": "To Kill a Mockingbird",
  "author": "Harper Lee",
  "Published_Year": 1960,
  "genres": ["Southern Gothic", "Drama"],
  "Pages": 281,
  "Available": true,
  "Rating": 4.6
},
{
  "title": "The Great Gatsby",
  "author": "F. Scott Fitzgerald",
  "Published_Year": 1925,
  "genres": ["Tragedy"],
  "Pages": 180,
  "Available": true,
  "Rating": 4.4
},
{
  "title": "Brave New World",
  "author": "Aldous Huxley",
  "Published_Year": 1932,
  "genres": ["Science Fiction", "Dystopian"],
  "Pages": 311,
  "Available": true,
  "Rating": 4.2
},
}
)
}
< {

```

Inserted Data:

Mongosh:

```

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('681c8ec5a00c09200c8db6a4'),
    '1': ObjectId('681c8ec5a00c09200c8db6a5'),
    '2': ObjectId('681c8ec5a00c09200c8db6a6'),
    '3': ObjectId('681c8ec5a00c09200c8db6a7'),
    '4': ObjectId('681c8ec5a00c09200c8db6a8')
  }
}

```

Inserted Data:

Compass

My Queries

CONNECTIONS (1)

Search connections

NewDB

- admin
- bookshop
 - books
- config
- local
- uni_DB

NewDB > bookshop > books

Documents | Aggregations | Schema | Indexes | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | EXPORT DATA | UPDATE | DELETE

25

```
{
  "_id": ObjectId("681c8ec5a0c09200c8db6a4"),
  "title": "The Hobbit",
  "author": "J.R.R.Tolkien",
  "Published_Year": 1917,
  "genres": Array (2),
  "Pages": 310,
  "Available": true,
  "Rating": 4.8
}
```

```
{
  "_id": ObjectId("681c8ec5a0c09200c8db6a5"),
  "title": "1984",
  "author": "George Orwell",
  "Published_Year": 1949,
  "genres": Array (2),
  "Pages": 328,
  "Available": true,
  "Rating": 4.7
}
```

```
{
  "_id": ObjectId("681c8ec5a0c09200c8db6a6"),
  "title": "To Kill a Mockingbird",
  "author": "Harper Lee",
  "Published_Year": 1960,
  "genres": Array (2),
  "Pages": 281,
  "Available": true,
  "Rating": 4.6
}
```

Compass

My Queries

CONNECTIONS (1)

Search connections

NewDB

- admin
- bookshop
 - books
- config
- local
- uni_DB

Welcome | mongosh:NewDB | books

NewDB > bookshop > books

Documents | Aggregations | Schema | Indexes | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | EXPORT DATA | UPDATE | DELETE

```
{
  "_id": ObjectId("681c8ec5a0c09200c8db6a6"),
  "title": "To Kill a Mockingbird",
  "author": "Harper Lee",
  "Published_Year": 1960,
  "genres": Array (2),
  "Pages": 281,
  "Available": true,
  "Rating": 4.6
}
```

```
{
  "_id": ObjectId("681c8ec5a0c09200c8db6a7"),
  "title": "The Great Gatsby",
  "author": "F. Scott Fitzgerald",
  "Published_Year": 1925,
  "genres": Array (1),
  "Pages": 180,
  "Available": true,
  "Rating": 4.4
}
```

```
{
  "_id": ObjectId("681c8ec5a0c09200c8db6a8"),
  "title": "Brave New World",
  "author": "Aldous Huxley",
  "Published_Year": 1932,
  "genres": Array (2),
  "Pages": 311,
  "Available": true,
  "Rating": 4.2
}
```

4. Find all books

Implementation:

```
}  
> db.books.find()
```

Output:

```
>_MONGOSH  
< {  
  _id: ObjectId('681c8ec5a00c09200c8db6a4'),  
  title: 'The Hobbit',  
  author: 'J.R.R.Tolkein',  
  Published_Year: 1917,  
  genres: [  
    'Fantasy',  
    'Adventure'  
  ],  
  Pages: 310,  
  Available: true,  
  Rating: 4.8  
}  
{  
  _id: ObjectId('681c8ec5a00c09200c8db6a5'),  
  title: '1984',  
  author: 'George Orwell',  
  Published_Year: 1949,  
  genres: [  
    'Dystopian',  
    'Political Fiction'  
  ],  
  Pages: 328,  
  Available: true,  
  Rating: 4.7  
}
```


>_MONGOSH

```
{
  _id: ObjectId('681c8ec5a00c09200c8db6a6'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  Published_Year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  Pages: 281,
  Available: true,
  Rating: 4.6
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a7'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  Published_Year: 1925,
  genres: [
    'Tragedy'
  ],
  Pages: 180,
  Available: true,
  Rating: 4.4
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
```

>_MONGOSH

```
  Available: true,
  Rating: 4.6
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a7'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  Published_Year: 1925,
  genres: [
    'Tragedy'
  ],
  Pages: 180,
  Available: true,
  Rating: 4.4
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  Published_Year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  Pages: 311,
  Available: true,
  Rating: 4.2
}
```

bookshop >

5. Find all books published after 1950

Implementation:

```
<  
> db.books.find({Published_Year:{$gt:1950}})  
< f
```

Output:

```
< {  
  _id: ObjectId('681c8ec5a00c09200c8db6a6'),  
  title: 'To Kill a Mockingbird',  
  author: 'Harper Lee',  
  Published_Year: 1960,  
  genres: [  
    'Southern Gothic',  
    'Drama'  
  ],  
  Pages: 281,  
  Available: true,  
  Rating: 4.6  
}
```

6. Find the available books.

Implementation:

```
.. }  
> db.books.find({Available:true})  
< {
```

Output:

```
< {  
  _id: ObjectId('681c8ec5a00c09200c8db6a4'),  
  title: 'The Hobbit',  
  author: 'J.R.R.Tolkein',  
  Published_Year: 1917,  
  genres: [  
    'Fantasy',  
    'Adventure'  
  ],  
  Pages: 310,  
  Available: true,  
  Rating: 4.8  
}  
{  
  _id: ObjectId('681c8ec5a00c09200c8db6a5'),  
  title: '1984',  
  author: 'George Orwell',  
  Published_Year: 1949,  
  genres: [  
    'Dystopian',  
    'Political Fiction'  
  ],  
  Pages: 328,  
  Available: true,  
  Rating: 4.7  
}
```

```
>_MONGOSH
{
  _id: ObjectId('681c8ec5a00c09200c8db6a6'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  Published_Year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  Pages: 281,
  Available: true,
  Rating: 4.6
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a7'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  Published_Year: 1925,
  genres: [
    'Tragedy'
  ],
  Pages: 180,
  Available: true,
  Rating: 4.4
}
```

```
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  Published_Year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  Pages: 311,
  Available: true,
  Rating: 4.2
}
```

7. Find all books that belong to the "Dystopian" genre.

Implementation:

```
<
> db.books.find({genres: "Dystopian"})
< {
```

Output:

```
< {
  _id: ObjectId('681c8ec5a00c09200c8db6a5'),
  title: '1984',
  author: 'George Orwell',
  Published_Year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  Pages: 328,
  Available: true,
  Rating: 4.7
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  Published_Year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  Pages: 311,
  Available: true,
  Rating: 4.2
}
```

8. Find the books which published after 1960 and had rating more than 4.0.

Implementation:

```
    Rating: 4.2  
  }  
> db.books.find({Published_Year:{$gt:1960}, Rating:{$gt:4.0}})  
<  
bookshop>
```

Output:

No outputs because there are no any books published after 1960.

9. Project only title and author fields:

Implementation:

```
> db.books.find({}, {title:1, author:1, _id:0})
```

Output:

```
< {
  title: 'The Hobbit',
  author: 'J.R.R.Tolkein'
}
{
  title: '1984',
  author: 'George Orwell'
}
{
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee'
}
{
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald'
}
{
  title: 'Brave New World',
  author: 'Aldous Huxley'
}
```

10. Update the book "1984" to set available: false:

Implementation:

```
}  
> db.books.updateOne  
  ({ "title": "1984" },  
    { $set: { "Available": false }  
  })
```

Output:

Mongosh:

```
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

Updated Data:

The screenshot shows the MongoDB Compass interface. On the left, the database structure is shown with 'local' and 'uni_DB' selected. The main panel displays a list of documents from the 'books' collection. The document for '1984' by George Orwell is highlighted, showing its updated state: 'Available' is now 'false'. The document for 'To Kill a Mockingbird' by Harper Lee is also visible, showing 'Available' as 'true'.

Document	Available	Rating
{ Available: true, Rating: 4.8 }	true	4.8
{ _id: ObjectId('681c8ec5a00c09200c8db6a5'), title: '1984', author: 'George Orwell', Published_Year: 1949, genres: Array (2), Pages: 328, Available: false, Rating: 4.7 }	false	4.7
{ _id: ObjectId('681c8ec5a00c09200c8db6a6'), title: 'To Kill a Mockingbird', author: 'Harper Lee', Published_Year: 1960, genres: Array (2), Pages: 281, Available: true, Rating: 4.6 }	true	4.6

11. Increase the rating of "Brave New World" by +0.1:

Implementation:

```
> db.books.updateOne(
  {"title": "Brave New World" },
  {$inc: {"Rating": 0.1}}
)
```

Output:

Mongosh:

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Increased Data:

```
_id: ObjectId('681c8ec5a00c09200c8db6a8')
title: "Brave New World"
author: "Aldous Huxley"
Published_Year: 1932
▶ genres: Array (2)
Pages: 311
Available: true
Rating: 4.3
```

12. Add a new genre "Classic" to "The Great Gatsby":

Implementation:

```
> db.books.updateOne(
  {"title": "The Great Gatsby"},
  {$push: {"genres": "Classic"}}
)
```

Output:

Mongosh:

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Updated Data:

The screenshot shows a MongoDB data viewer interface. At the top, there's a toolbar with icons for adding, editing, deleting, and other actions. Below the toolbar, a list of documents is displayed. The first document is expanded, showing its fields: `_id` (ObjectId), `title` ("The Great Gatsby"), `author` ("F. Scott Fitzgerald"), `Published_Year` (1925), `genres` (Array of 2: "Tragedy", "Classic"), `Pages` (180), `Available` (true), and `Rating` (4.4). The second document is also visible, showing `title` ("Brave New World"), `author` ("Aldous Huxley"), `Published_Year` (1932), `genres` (Array of 2), `Pages` (311), and `Available` (true).

13.Sort books by published year ascending:

Implementation:

```
> db.books.find().sort({Published_Year:1})
```

Output:

```
>_MONGOOSH
> db.books.find().sort({Published_Year:1})
< {
  _id: ObjectId('681c8ec5a00c09200c8db6a4'),
  title: 'The Hobbit',
  author: 'J.R.R.Tolkien',
  Published_Year: 1917,
  genres: [
    'Fantasy',
    'Adventure'
  ],
  Pages: 310,
  Available: true,
  Rating: 4.8
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a7'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  Published_Year: 1925,
  genres: [
    'Tragedy',
    'Classic'
  ],
  Pages: 180,
  Available: true,
  Rating: 4.4
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
```

```
>_MONGOOSH
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  Published_Year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  Pages: 311,
  Available: true,
  Rating: 4.3
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a5'),
  title: '1984',
  author: 'George Orwell',
  Published_Year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  Pages: 328,
  Available: false,
  Rating: 4.7
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a6'),
  title: 'To Kill a Mockingbird',
```

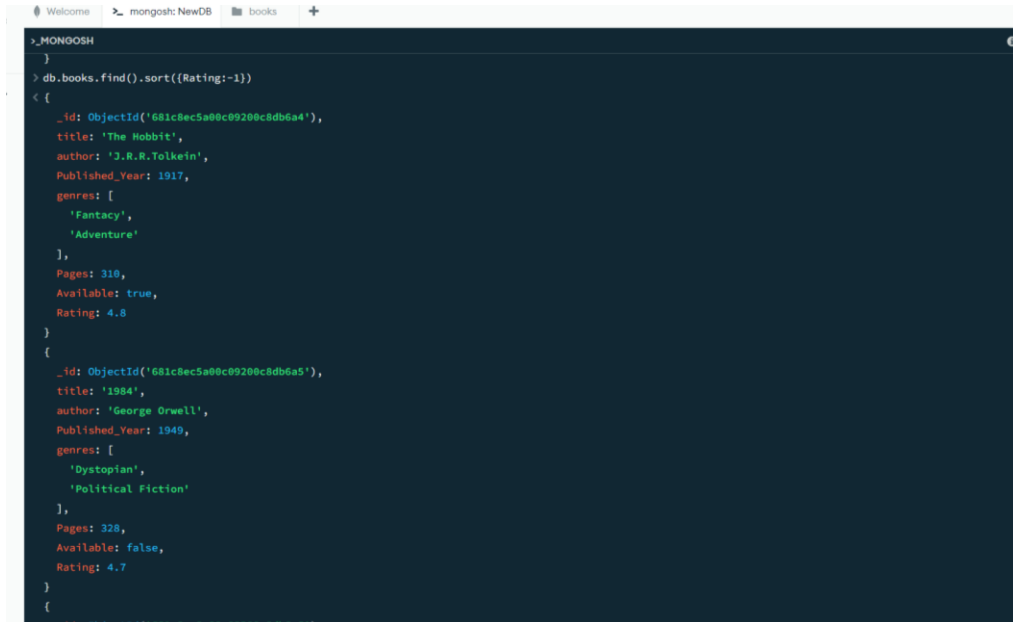
```
>_MONGOOSH
Rating: 4.3
}
{
  _id: ObjectId('681c8ec5a0c09200c8db6a5'),
  title: '1984',
  author: 'George Orwell',
  Published_Year: 1949,
  genres: [
    'Dystopian',
    'Political Fiction'
  ],
  Pages: 328,
  Available: false,
  Rating: 4.7
}
{
  _id: ObjectId('681c8ec5a0c09200c8db6a6'),
  title: 'To Kill a Mockingbird',
  author: 'Harper Lee',
  Published_Year: 1960,
  genres: [
    'Southern Gothic',
    'Drama'
  ],
  Pages: 281,
  Available: true,
  Rating: 4.6
}
bookshop >
```

14.Sort books by rating descending:

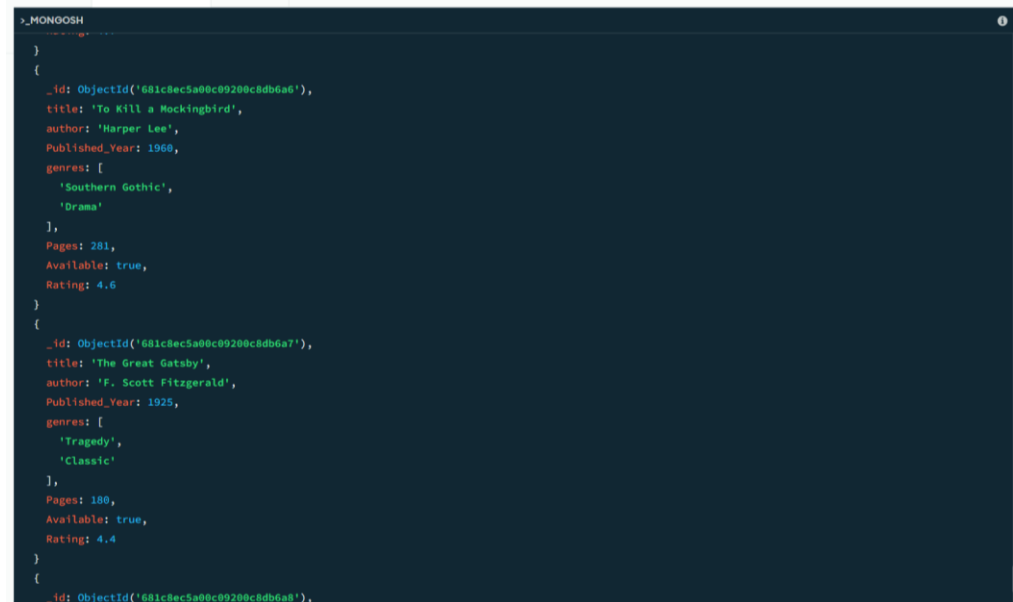
Implementation:

```
> db.books.find().sort({Rating:-1})
```

Output:



```
> db.books.find().sort({Rating:-1})
{
  "_id": ObjectId("681c8ec5a00c09200c8db6a4"),
  "title": "The Hobbit",
  "author": "J.R.R.Tolkien",
  "Published_Year": 1917,
  "genres": [
    "Fantasy",
    "Adventure"
  ],
  "Pages": 310,
  "Available": true,
  "Rating": 4.8
}
{
  "_id": ObjectId("681c8ec5a00c09200c8db6a5"),
  "title": "1984",
  "author": "George Orwell",
  "Published_Year": 1949,
  "genres": [
    "Dystopian",
    "Political Fiction"
  ],
  "Pages": 328,
  "Available": false,
  "Rating": 4.7
}
```



```
{
  "_id": ObjectId("681c8ec5a00c09200c8db6a6"),
  "title": "To Kill a Mockingbird",
  "author": "Harper Lee",
  "Published_Year": 1960,
  "genres": [
    "Southern Gothic",
    "Drama"
  ],
  "Pages": 281,
  "Available": true,
  "Rating": 4.6
}
{
  "_id": ObjectId("681c8ec5a00c09200c8db6a7"),
  "title": "The Great Gatsby",
  "author": "F. Scott Fitzgerald",
  "Published_Year": 1925,
  "genres": [
    "Tragedy",
    "Classic"
  ],
  "Pages": 180,
  "Available": true,
  "Rating": 4.4
}
{
  "_id": ObjectId("681c8ec5a00c09200c8db6a8"),
```

```
>_MONGOSH
  Rating: 4.6
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a7'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  Published_Year: 1925,
  genres: [
    'Tragedy',
    'Classic'
  ],
  Pages: 180,
  Available: true,
  Rating: 4.4
}
{
  _id: ObjectId('681c8ec5a00c09200c8db6a8'),
  title: 'Brave New World',
  author: 'Aldous Huxley',
  Published_Year: 1932,
  genres: [
    'Science Fiction',
    'Dystopian'
  ],
  Pages: 311,
  Available: true,
  Rating: 4.3
}
```


15.Delete all books with a rating lower than 4.5:

Implementation:

```
> db.books.deleteMany({"Rating":{"$lt:4.5}})
```

Output:

Mongosh:

```
< {  
  acknowledged: true,  
  deletedCount: 2  
}
```

Deleted:

Documents 1

Aggregations

Schema

Indexes 1

Validation

🕒

⌵

{}

Project

{ field: 0 }

Sort

{ field: -1 } or [['field', -1]]

Collation

{ locale: 'simple' }

Index Hint

{ field: -1 }

Generate query ⚡

Explain

Reset

Find

⌵

Options ⌵

Max Time MS

60000

Skip

0

Limit

0

➕ ADD DATA ⌵

📄 EXPORT DATA ⌵

✎ UPDATE

🗑 DELETE

25 ⌵

1 - 3 of 3

↺

↻

⌵

⌵

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🏠 books

	_id ObjectId	title String	author String	Published_Year Int32	genres Array	Pages Int	
1	ObjectId('681c8ec5a00c092...	"The Hobbit"	"J.R.R.Tolkein"	1917	[] 2 elements	310	✎ 🗑 📄
2	ObjectId('681c8ec5a00c092...	"1984"	"George Orwell"	1949	[] 2 elements	328	✎ 🗑 📄
3	ObjectId('681c8ec5a00c092...	"To Kill a Mockingbird"	"Harper Lee"	1960	[] 2 elements	281	✎ 🗑 📄